

THROMBOSIS AND HEMOSTASIS

Oxidized PDI promotes thrombus formation in oxidative stress

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KEY POINTS

- The ability of PDI to oxidize substrates drives thrombus formation *in vivo*.
- PDI promotes thrombus formation in the setting of oxidative stress through sulfenylation via a peroxiredoxin-like mechanism.

Protein disulfide isomerase (PDI) functions in thrombus formation *in vivo* and represents a viable target for antithrombotic therapy. PDI is a redox sensor that can either reduce or oxidize substrates depending on the redox environment. However, whether PDI functions primarily as a reductase or an oxidase in the context of thrombus formation remains unknown. Here, we used pharmacological and genetic approaches along with PDI mutants to determine how the PDI redox state affects thrombus formation. LOC14, which inhibits PDI reductase activity and induces PDI oxidation, promoted thrombus formation in arteries exposed to ferric chloride and enhanced laser injury-induced platelet accumulation and fibrin formation in cremaster arterioles. Blocking antibodies targeting PDI reversed the prothrombotic effect of LOC14. Evaluation of sulfenylation-mediated PDI oxidation using the C53A, C56A, R120D, and T101A PDI mutants showed that the sulfenylation mechanism of PDI resembles that of hydrogen peroxide

(H₂O₂) reduction by peroxiredoxins. These studies identified PDI mutants that failed to undergo H₂O₂-mediated oxidation, but showed normal reductase activity. When tested *in vivo*, either wild-type PDI or the R120D mutant fully restored normal thrombus formation following morpholino-induced knockdown of PDI or in mice with platelet-specific knockout of PDI. In contrast, the sulfenylation-impaired R120D mutant PDI was unable to fully restore thrombus formation in the setting of oxidative stress induced in mice by genetic deletion of glutathione peroxidase 3 or by infusion of oxidized low-density lipoproteins. These studies show that PDI-catalyzed oxidation drives thrombosis and demonstrates a mechanism of peroxide-mediated oxidation of PDI that contributes to the prothrombotic response of oxidative stress.

Introduction

Thrombosis most commonly occurs under conditions of oxidative stress.¹ Aging, inflammation, hyperlipidemia, diabetes, hypertension, and malignancy are all associated with both oxidative stress and thrombosis.²⁻⁴ Production of reactive oxygen species with oxidation of lipids and proteins coupled with impaired antioxidant defenses are common among these conditions. Antioxidants such as vitamin C and E have had limited impact in modifying thrombotic outcomes in these disease settings,⁵ an observation attributed to both the capacity of endogenous and pathological oxidant enzyme systems to overwhelm oxidant scavengers and the rapid kinetics and high local concentrations that oxidants achieve.⁶ In

this regard, strategies targeting the enzymes and oxidative pathways that mediate thrombotic complications of oxidative stress may be more effective than stoichiometric scavenging.

Protein disulfide isomerase (PDI) is a thiol isomerase that performs oxidative protein folding in the endoplasmic reticulum.⁷ Stimulation of vascular cells can release PDI into the extracellular environment,^{8,9} where it can promote thrombus formation. Inhibition of PDI blocks thrombus formation in several animal models of thrombosis¹⁰⁻¹³ and clinical studies show that PDI may be a viable antithrombotic target.¹⁴ Evaluation of thrombus formation in mice expressing catalytic cysteine PDI mutants confirmed that the oxidoreductase activity of PDI is essential for its role in thrombosis.¹⁵ However, whether PDI

functions primarily as reductase or oxidase in the context of thrombus formation is largely unstudied.

In its role as an oxidoreductase, PDI can both cleave and form disulfide bonds. Cleavage of disulfides by PDI is typically achieved through nucleophilic attack by an N-terminal free thiolate of the Cys-Gly-His-Cys (CGHC) catalytic motif reacting with the disulfide bond of the substrate.¹⁶ This reaction results in a transient mixed disulfide bond between PDI and its substrate, which resolves with reduction of the substrate disulfide and formation of a disulfide within the PDI catalytic motif. Oxidation of substrates, which is initiated by nucleophilic attack of a free thiolate in the substrate, also proceeds through a mixed disulfide intermediate (supplemental Figure 1, available on the *Blood* website).¹⁷ The formation of mixed disulfides has been leveraged to identify substrates of extracellular PDI in platelets and plasma.¹⁸⁻²² A study comparing preoxidized and prereduced PDI trapping mutants showed different patterns of substrate interactions depending on the redox state of the trapping mutant.²²

Peroxides are central mediators of oxidative stress and play a prominent role in driving prothrombotic responses in this setting. Hydrogen peroxide (H₂O₂) promotes platelet hyper-reactivity and thrombosis.²³⁻²⁶ Vascular peroxides including inorganic peroxides (eg, H₂O₂) and lipid peroxidases can elicit direct modification of amino acid side chains, including oxidative cysteine modifications, of coagulation-related proteins.^{27,28} Glutathione peroxidases (GPx) are a large family of antioxidant enzymes that reduce peroxides. Gene variants in GPx family enzymes are associated with stroke, myocardial infarction, and peripheral arterial disease.²⁹⁻³⁴ Peroxide-driven oxidation of free thiols within extracellular thiol isomerases has recently been invoked as a mechanism for promoting thrombus formation.^{35,36} We have previously shown that peroxides promote PDI oxidase activity by a cysteine sulfenylated intermediate.³⁶ Sulfenylation is the first oxidative cysteine modification of the cysteine thiol by peroxides before further sulfur oxoform formation (eg, sulfinic and sulfonic acids, disulfides). However, the molecular mechanisms by which peroxides induce PDI oxidation have not previously been evaluated.

We found that a compound that inhibits PDI reductase activity but promotes its oxidase activity, enhances thrombus formation in vivo. Studies evaluating the mechanism of sulfenylation-mediated oxidation of PDI showed that it is sulfenylated by a peroxiredoxin-like mechanism and led to the identification of mutants that can be oxidized by oxidized glutathione (GSSG) but not H₂O₂. Evaluation of these sulfenylation-deficient PDI mutants in vivo demonstrated that sulfenylated PDI contributes to thrombus formation in the presence but not the absence of oxidative stress. Thus, these studies demonstrate that the oxidase activity of PDI drives thrombus formation in vivo and indicate different pathways for oxidation of PDI.

Methods

Expression and purification of recombinant human PDI

Wild-type (WT) and variant human PDIs were expressed and purified as previously described³⁶ and further detailed in the supplemental Data.

Cysteine sulfenylation detection

Sulfenylation detection was performed as previously described³⁶ and further detailed in the supplemental Data.

Protein labeling and confocal single-molecule FRET experiments

Site-specific labeling at positions 88 and 467 with the Förster resonance energy transfer (FRET) pair Atto-550/Atto-647N was achieved by click chemistry, as detailed in previous publications.³⁷⁻⁴⁰ Further details of single-molecule FRET experiments are described in the supplemental Data.

Laser-injury cremaster arteriole thrombosis model

Thrombus formation in response to laser injury was measured as described previously.^{36,41} Briefly, cremaster muscle arterioles were injured using a MicroPoint Laser System (Andor, Belfast, United Kingdom). Platelet and fibrin accumulation were measured by infusion of DyLight 488- or 649-labeled antiplatelet glycoprotein Ib β (CD42c; 0.1 μ g/g body weight; Emfret Analytics) and DyLight 488- or 649-labeled anti-fibrin (clone 59D8; 0.3 μ g/g) antibodies through a jugular vein catheter. For rescue studies, 200 μ g of recombinant WT or variant PDI were infused through the catheter. Data were collected before and after laser injury using the brightfield, 488/520 nm and 640/670 nm channels. Images were captured for 250 seconds at 2 frames/s using a charge-coupled device (CCD) camera (ORCA Flash 4.0, Hamamatsu Photonics, Iwata, Japan) on an AX-70 upright fluorescence microscope (Olympus, Tokyo, Japan). The area under the curve was calculated for individual thrombi and normalized to injury lengths to evaluate statistical significance. Injury lengths were determined as previously described.⁴¹

FeCl₃-mediated carotid artery occlusion thrombosis model

Ferric chloride (FeCl₃)-mediated thrombus formation in the carotid artery was performed as previously described.⁴² C57Bl/6J WT, GPx3^{-/-}, PDI^{fl/fl}.PF4 cre⁻, or PDI^{fl/fl}.PF4 cre⁺ mice, aged 8 to 10 weeks, were anesthetized with a cocktail of 125 mg/kg ketamine and 12 mg/kg xylazine. The left jugular vein was cannulated for infusion of a DyLight 649 conjugated antibody against glycoprotein Ib β (Emfret Analytics; 0.1 μ g/g) and 50 mg/kg pentobarbital for maintenance anesthetics. The right carotid artery was isolated with a piece of nonfluorescent plastic placed under the vessel. Vessel autofluorescence (excitation wavelength 488 nm, 50 milliseconds) and platelet accumulation (excitation 650 nm, 50 milliseconds) were captured for 15 seconds (2 frames/s) every minute up to 30 minutes. In some studies, before injury, the mouse received IV administration of 20 mg/kg lead optimization compound (LOC)14, oxy-LOC14, vehicle control, 2.5 mg/kg oxidized low-density lipoprotein (oxLDL), or 200 μ g of recombinant WT, R120D, or catalytically inactive dead mutant (DM) PDI for 5 minutes. The injury was made by topical application of a 1 mm $\times$ 2 mm filter paper saturated with 10% FeCl₃ for 3 minutes. Thrombi formation was observed using a 10 $\times$ 0.3 normal aperture (NA) water immersion objective mounted to an AX-70 fluorescence microscope (Olympus) equipped with a CCD camera (ORCA Flash 4.0, Hamamatsu Photonics) or a 20 $\times$ 1.0 NA water immersion objective mounted to a Zeiss Axio Examiner.Z1 equipped with a complementary metal-oxide-semiconductor camera

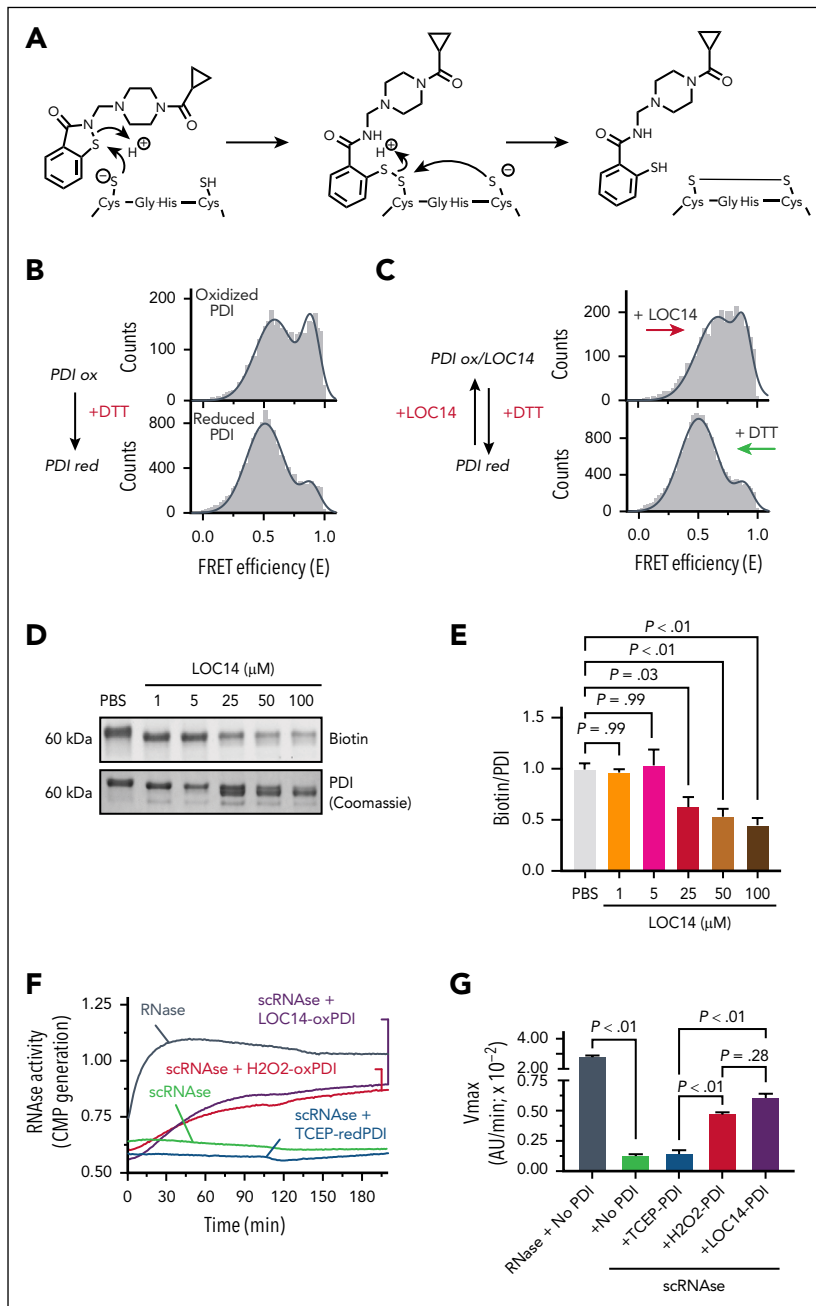


Figure 1. LOC14 oxidizes PDI and promotes its oxidase activity. (A) Mechanism of LOC14 oxidizing the catalytic CGHC motif of PDI. (B) FRET histograms show PDI 88/467 (80 pM) in the absence (oxidized PDI) and presence of 200 μ M dithiothreitol (DTT) (reduced PDI), followed by (C) the addition of 750 μ M LOC14, and 2 mM DTT. LOC14 causes the conformational equilibrium to shift toward higher FRET, but adding DTT reverses this effect. This indicates that LOC14 binding favors conformational states resembling those seen by single-molecule FRET in oxidized PDI, and that the conformational changes induced by LOC14 are reversible. (D) Western blot to detect the loss of free thiols on PDI by measuring maleimide-PEG2-biotin after incubating PDI with increasing concentration of LOC14. (E) Quantitation of western blot studies shown in panel D ($n = 3$). (F) Kinetic tracings of cytidine 3',5'-monophosphate (CMP) formation from cyclic CMP hydrolysis of RNase derived by measuring absorbance at A295 in the absence and presence of PDI with indicated conditions. (G) $V_{\max}$ quantification of activity of RNase exposed to reduced vs oxidized PDI. $N = 3$ independent experiments in panels B-F. PBS, phosphate-buffered saline; PDI ox, oxidized PDI; PDI red, reduced PDI; $V_{\max}$, maximal velocity.

(ORCA Fusion BT, Hamamatsu Photonics). Data were analyzed by Slidebook 6.0 (Intelligent Imaging Innovations).

Other experiments are detailed in the supplemental Methods.

Results

LOC14 inhibits PDI reductase activity and promotes its oxidase activity

A major difficulty in distinguishing between *in vivo* activities of oxidized PDI and reduced PDI is that, upon introduction into the circulation, the oxidation state of PDI changes to reflect the prevailing redox conditions within the bloodstream. To

constrain PDI to an oxidized state, we used LOC14, a benzothiazole oxidizer of PDI discovered during a high throughput screen designed to identify inhibitors of PDI reductase activity.⁴³ LOC14 was found to be a reversible inhibitor of PDI reductase activity that binds to the "a" domain with nanomolar affinity.⁴³ The proposed mechanism of LOC14 inhibition of PDI is by oxidation of active site cysteines, forcing the enzyme into an oxidized conformation (Figure 1A). The conformation of PDI is highly sensitive to redox state.^{38,39} To evaluate the premise that LOC14 stabilizes oxidized PDI and promotes an oxidized conformation, we used a modified PDI construct containing fluorescent labels at amino acids 88 and 467, and evaluated the effect of LOC14 on PDI conformation using enhanced FRET.^{38,39,44} Ambient oxidation of PDI promoted efficient

FRET, which was reversed when the reducing agent dithiothreitol was added (Figure 1B). Incubation with LOC14 induced a tracing that closely resembles that of oxidized PDI rather than that of reduced PDI (Figure 1C).

We also assessed the effect of LOC14 on the active sites of PDI. To directly determine whether LOC14 decreases availability of reactive thiols, we exposed PDI to different concentrations of LOC14 and then tested for its ability to bind maleimide–polyethylene glycol biotin. Maleimide–polyethylene glycol biotin binding decreased with increasing concentrations of LOC14 (Figure 1D–E), indicating that incubation with LOC14 reduced free thiols in PDI. Although LOC14 inhibits reductase activity, its oxidation of PDI (Figure 1A) could potentially enhance the ability of PDI to function as an oxidase and isomerase. To evaluate this possibility, we tested LOC14 in an RNase refolding assay. In this assay, ribonuclease (RNase) is denatured and reduced (scrambled RNase) and then subsequently incubated with either reduced or oxidized PDI. Oxidative renaturing of scrambled RNase from PDI's oxidase activity promotes functional RNase hydrolysis of cyclic cytidine 3',5'-monophosphate to cytidine 3',5'-monophosphate. No activity was recovered upon incubation with reduced PDI (Figure 1F–G). However, oxidation of PDI by either H₂O₂ or LOC14 enabled it to refold RNase into an active enzyme. These studies show that LOC14 inhibits PDI reductase activity, but stimulates PDI oxidase activity *in vitro*.

LOC14 promotes thrombus formation

To determine whether reduced or oxidized PDI is more important for thrombus formation *in vivo*, we analyzed the effect of LOC14 on platelet accumulation and fibrin formation in mice following laser-induced vascular injury of cremaster arterioles. Since LOC14 inhibits the activity of reduced PDI, we reasoned that if the reduced state of PDI contributes directly to thrombus formation, LOC14 would inhibit thrombus formation. Conversely, if the oxidized state of PDI contributed directly to thrombus formation, LOC14 would promote thrombus formation. To verify that any observed effects of LOC14 on thrombus formation are secondary to PDI, we evaluated LOC14 in the presence of either nonimmune immunoglobulin G or anti-PDI antibody (RL90). In the absence of anti-PDI antibody (nonimmune immunoglobulin G only), LOC14 augmented platelet accumulation by 3.5-fold (Figure 2A–B,D) and fibrin formation by 4.9-fold (Figure 2A,C,E). Anti-PDI antibody blocked both LOC14-augmented platelet accumulation and fibrin formation in this model (Figure 2A–E; supplemental Videos 1–4), confirming the role of PDI in platelet accumulation and fibrin formation as previously shown by our group and others.^{10,45} Oxy-LOC14⁴³ is an analog of LOC14 that differs in a single atom substitution of oxygen for the critical sulfur atom (supplemental Figure 2A) that promotes nucleophilic attack of LOC14 by C53.⁴³ In contrast to LOC14, oxy-LOC14 did not significantly affect platelet accumulation or fibrin formation following laser injury (supplemental Figure 2B–C; supplemental Videos 5 and 6). LOC14 also decreased occlusion time following exposure of the right carotid arteries to FeCl₃ injury relative to vehicle treated mice (Figure 2F–G). In contrast, oxy-LOC14 had no effect, further indicating that oxidized PDI promotes thrombus formation *in vivo*. We further tested oxidized PDI on tail bleeding hemostasis. LOC14 did not

significantly affect tail bleeding (supplemental Figure 2D,E). These studies show that LOC14 promotes thrombus formation and that it does so in a PDI-dependent manner.

Peroxide-mediated oxidation of PDI

A 2-electron oxidant is a chemical species that pulls 2 electrons away from a target molecule in a single step, thereby oxidizing the target. PDI favors reactions with 2-electron oxidants⁴⁶ and can interact with many such species including H₂O₂, urate hydroperoxide, peroxyxynitrite, and glutathione disulfide. Our previous work demonstrated that peroxide-induced oxidation of PDI proceeds via a sulfenylated intermediate (supplemental Figure 3). Sulfenylation of PDI occurs primarily within the “a” domain,³⁶ suggesting that PDI may function as a peroxiredoxin-like enzyme. Like PDI, peroxiredoxins strongly favor reactions with 2-electron oxidants.^{47,48} Peroxiredoxins contain a conserved catalytic center with a peroxidatic cysteine that functions along with adjacent arginine and threonine residues that stabilize the thiolate form and position the peroxide for nucleophilic attack (supplemental Figure 4). Structural comparison between peroxiredoxin 5⁴⁹ (Figure 3A) and the “a” domain of PDI⁵⁰ (Figure 3B) revealed that the alignment of key catalytic residues such as T44, C47, and R127 in peroxiredoxin 5 correspond to C53, C56, and R120 in PDI (Figure 3C). These residues, along with T101 (adjacent to C56), are highly conserved across PDI superfamily members (Figure 3D; supplemental Figure 5).

The chemical probe BTB (1-(4-Pentyn-1-yl)-1H-2,1-benzothiazin-4(3H)-one 2,2-dioxide) selectively reacts with sulfenylated cysteines to form stable thioether adducts detectable by azide-alkyne cycloaddition (supplemental Figure 6). Analysis of PDI mutants C56A and C53A (Figure 3E), as well as C400A and C56A/C400A (supplemental Figure 7) showed that C56A alone abolished BTB reactivity, indicating C56 is necessary and sufficient for H₂O₂-induced sulfenylation. In contrast, C53A had minimal effect. Mutants R120D and T101A showed significantly reduced BTB labeling, implicating these residues as key participants in peroxide-mediated oxidation (Figure 3E–F). Mutations of A58G, A60G, and A58G/A60G designed to interfere with the macrodipole of the α -helix proximal to the PDI active site motif did not affect sulfenylation (supplemental Figure 8). These observations indicate the importance of C56, R120, and T101 in PDI sulfenylation.

To validate these findings, we employed a fluorescence-based assay using a synthetic peptide that quenches upon disulfide bond formation. WT and mutant PDIs were treated with H₂O₂ to initiate PDI-catalyzed oxidation of the peptide probe. Fluorescence quenching resulting from disulfide bond formation was monitored in real time (Figure 3G). We have previously shown that oxidation of PDI by H₂O₂ occurs through a sulfenylated intermediate that resolves into a disulfide (supplemental Figure 3).³⁶ Consistent with previous results, R120D and T101A mutants exhibited significantly reduced initial quenching rates similar to inactive control (Figure 3H), indicating impaired sulfenylation and inefficient H₂O₂-dependent oxidation.

Although peroxides are key mediators of oxidative stress, the reduced glutathione (GSH)-oxidized glutathione (GSSG)

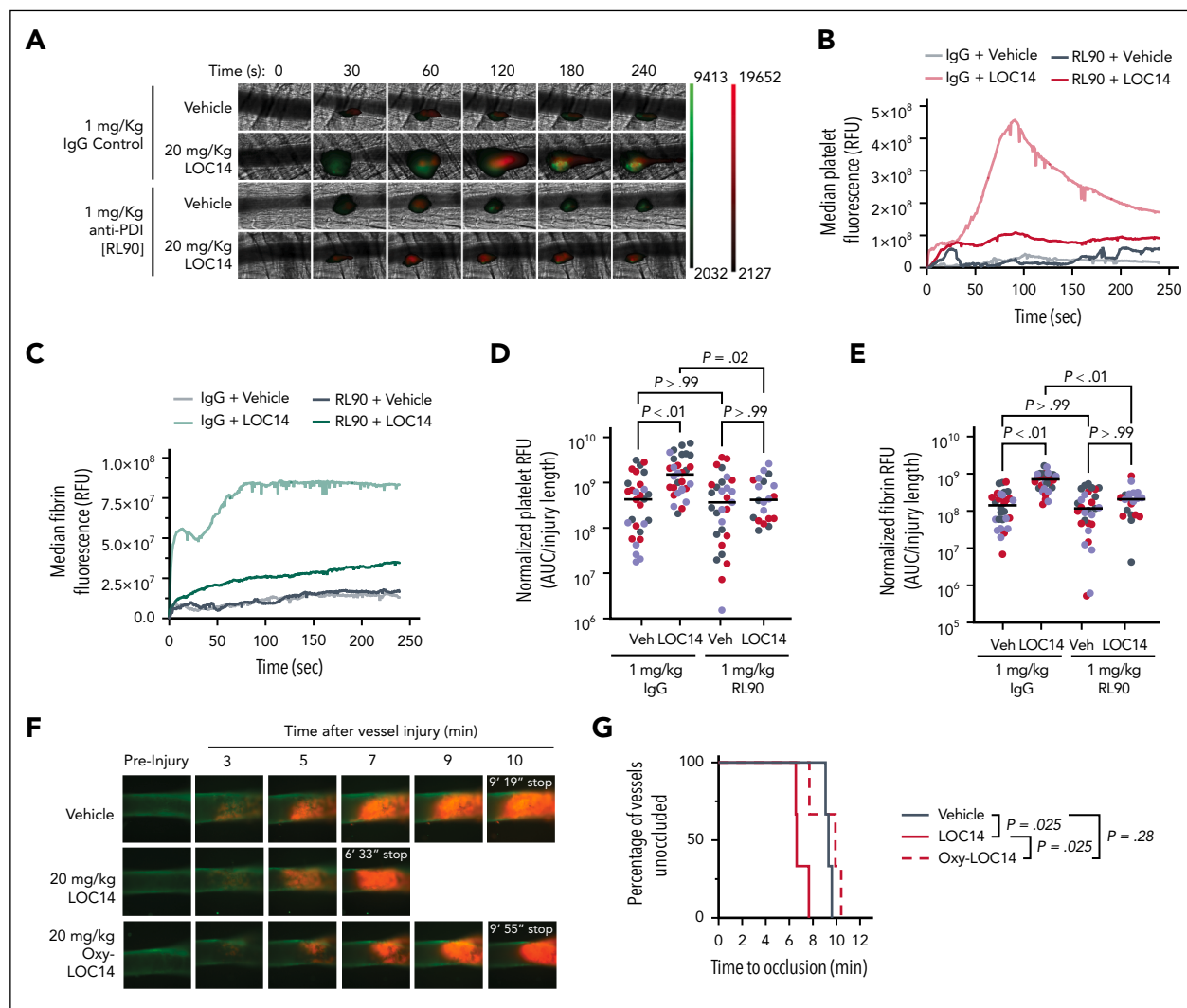


Figure 2. LOC14 promotes thrombus formation in a PDI-dependent manner. (A) Intravital video microscopy images of platelet (red) and fibrin (green) fluorescence by LOC14 after pretreatment with an anti-PDI blocking monoclonal or control immunoglobulin G (IgG) antibody. Kinetic tracings of the median platelet (B) or fibrin (C) fluorescence of panel A. Quantification of platelet (D) and fibrin (E) accumulation in the laser-injury thrombosis model showing that RL90 prevents the augmented thrombosis observed following LOC14-treatment. N = 30 injuries in IgG + vehicle treatment; N = 29 injuries in RL90 + Vehicle treatment; N = 30 IgG + LOC14 treatment; and N = 20, RL90 + LOC14 treatment. In each cohort of injuries, the injuries were from 3 different mice. Different colors represent injuries from different mice. (F) Intravital microscopy images of the FeCl₃-mediated carotid artery thrombosis model when vehicle (dimethyl sulfoxide), 20 μ g/g LOC14, or 20 μ g/g oxy-LOC14 were infused through IV. (G) Kaplan-Meier analysis of the percentage of vessels that remained unoccluded from panel F. N = 3 different mice/treatment in panels F-G. RFU, relative fluorescence unit.

system also plays a central role in redox regulation. Unlike peroxide-mediated sulfenylation, GSSG oxidizes PDI via thiol-disulfide exchange, reducing GSSG to GSH.⁵¹ To assess the role of R120 and T101 in this pathway, we used a di-eosin-GSSG assay.⁵² Both R120D and T101A mutants showed normal activity (Figure 3I), indicating that these residues are required for H₂O₂-mediated oxidation, but not for GSSG-dependent disulfide exchange.

Role of peroxide-mediated oxidation of PDI in thrombus formation

The identification of sulfenylation-defective PDI mutants enabled us to evaluate the role of PDI sulfenylation in thrombus formation in vivo. Since genetic knockdown of PDI results in embryonic lethality, endogenous PDI was knocked down using cell-penetrating morpholinos (MO).⁵³ Two

different antisense oligomer MO were used, one targeting the start codon adenine-uracil-guanine (AUG) and the other targeting the 5' untranslated region (5' UTR) of the PDI mRNA. Knockdown of PDI occurred over a period of 4 days, only occurred with targeted MO, and was selective for PDI over ERp5 (Figure 4A-C). On day 4 following knockdown of endogenous PDI, either WT or R120D mutant PDIs were infused into the mice. Median platelet accumulation during thrombus formation in mice exposed to a PDI targeted anti-sense oligomer MO (PDI AUG-MO) was decreased to 23% ($P < .05$) of that observed in mice exposed to a control MO (Figure 4D-F; supplemental Videos 7-11). Similar decreases were observed with a MO targeting the 5' UTR of the PDI transcript (PDI 5' UTR MO). Median fibrin formation was decreased to 36% ($P < .01$) following infusion of PDI AUG-MO and 47% ($P < .01$) following infusion of PDI 5' UTR MO

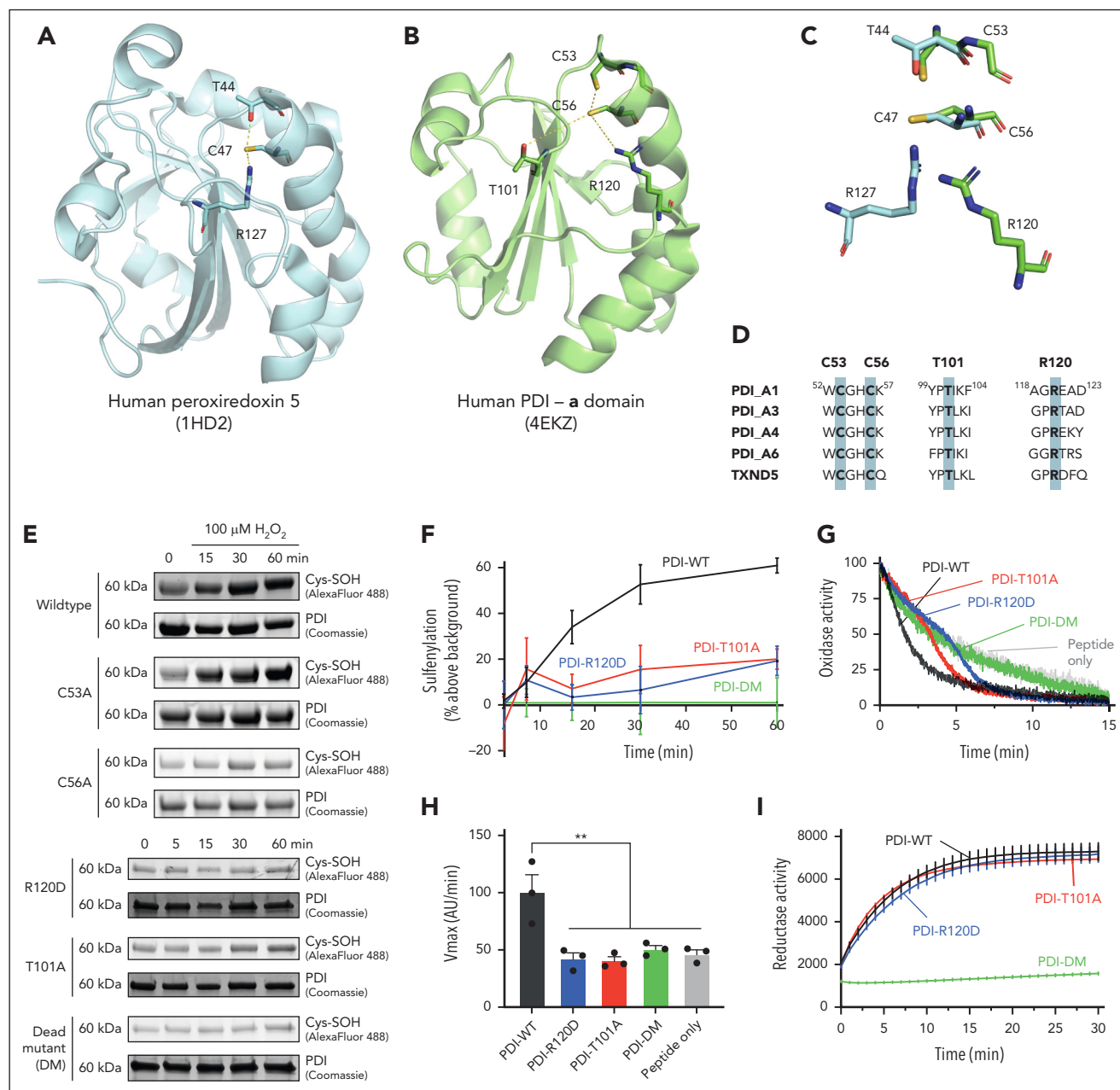


Figure 3. Residues R120 and T101 facilitate PDI sulfenylation, but are not essential for PDI reductase activity. (A) Structure of the catalytic domain of peroxiredoxin 5 highlighting the peroxidatic cysteine (C47) and key catalytic residues, R127 and T44. (B) Structure of the “a” domain of PDI highlighting residues (R120, T101, and C53) proposed to influence the thiolate form of C56. (C) Comparison between key residues in active-site of peroxiredoxin 5 (C47, R127, T44) and corresponding residues in PDI (C56, R120, C53). (D) Sequence alignment of PDI family members known to function in thrombus formation, highlighting residues analogous to PDI C53, C56, T101, and R120. (E) Detection of H_2O_2 -induced cysteine sulfenylation in recombinant WT PDI, C53A, C56A, R120D, T101A, or catalytically dead mutant (DM) (lacking active site cysteines). (F) Sulfenylation is deficient in an R120D, T101A, and a catalytic DM, wherein all the active site cysteines were mutated to alanines. (G) Kinetic tracings of the rate of disulfide transferring PDI oxidase activity by H_2O_2 between the WT (PDI-WT), sulfenylation-deficient PDI-R120D, and PDI-T101A, and the catalytically inactive DM (PDI-DM). (H) Quantification of the initial rates of oxidation from panel G. (I) Kinetic tracings of the rate of GSSG reduction between the sulfenylation-deficient R120D and T101A, DM, and WT in the reductase activity assay. N = 3 independent experiments in panels E-G. ** $P < .01$. Vmax, maximal velocity.

compared with control (Figure 4D,G-H). Infusion of either native PDI or the R120D mutant PDI reversed the defect in both platelet accumulation and fibrin formation, to the same extent (Figure 4). Thus, in WT mice with no obvious source of oxidative stress, the R120D PDI mutant functions normally to support thrombus formation following PDI knockdown.

To assess the role of H_2O_2 -mediated PDI oxidation in the setting of systemic oxidative stress, we used mice that lack GPx3. GPx3 serves a critical role in catalyzing the reduction of H_2O_2 to water and oxygen. Variants in GPx3 are associated with thrombosis in humans.^{30,32} GPx3^{-/-} mice are phenotypically normal, but have elevated endogenous H_2O_2 , increased

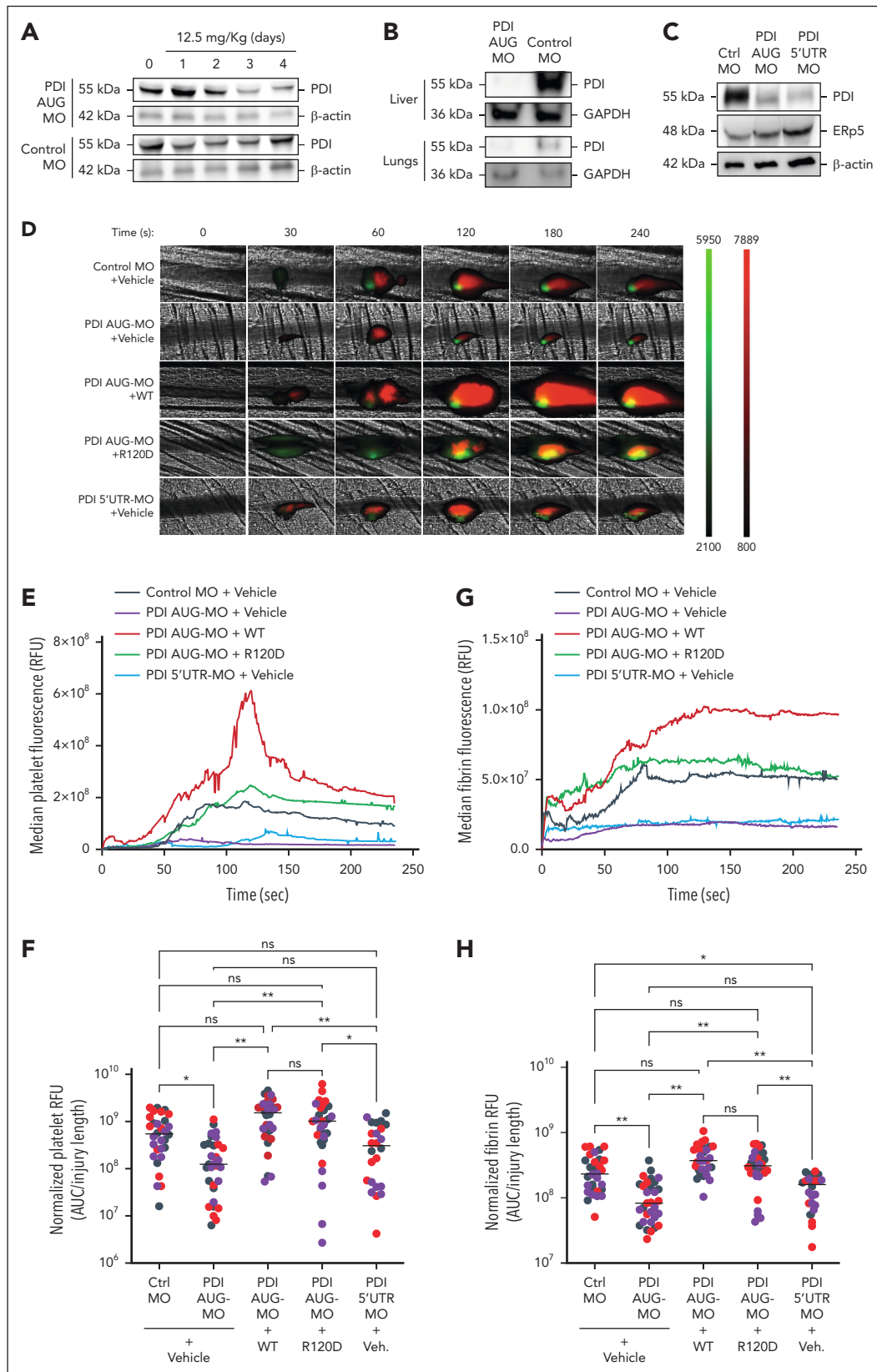
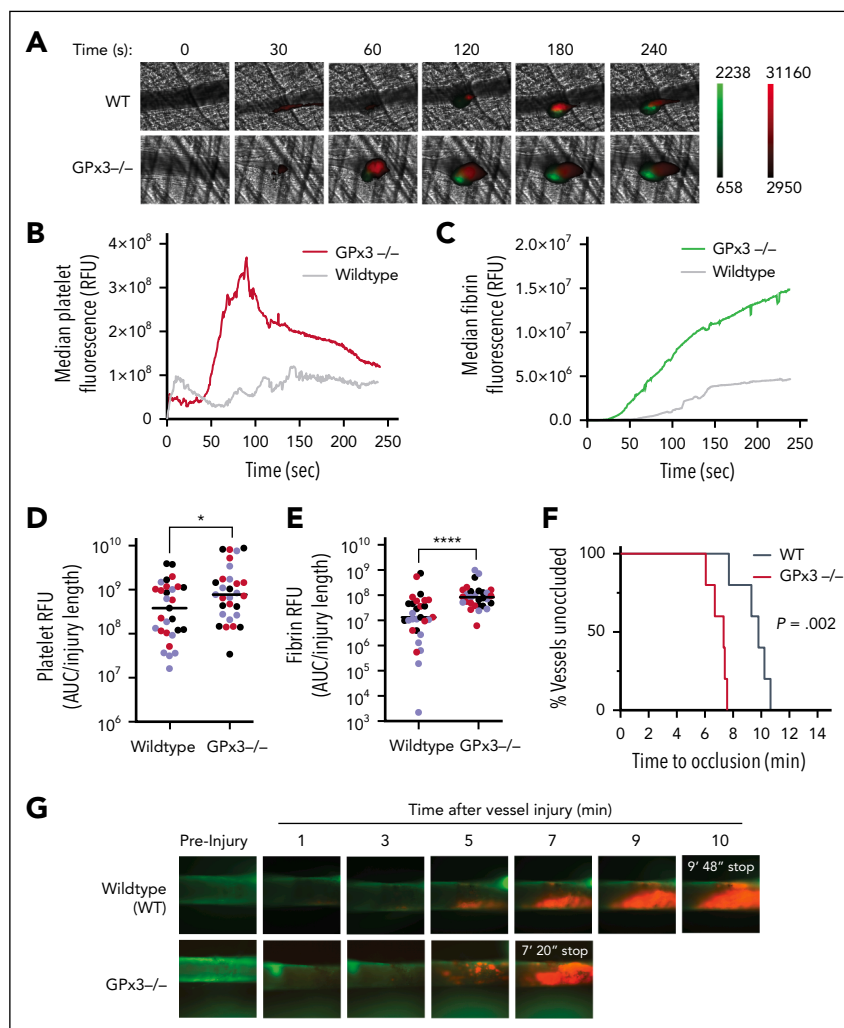


Figure 4. WT and sulfenylation-deficient R120D restores thrombus formation following MO-mediated knockdown of PDI in mice. (A) Western blot for PDI expression in platelets following 12.5 mg/kg IV treatment of PDI AUG MO or control MO knockdown experiments. (B) Western blot for PDI expression in the liver and lungs of PDI AUG MO or control MO treated mice at day 4 of treatment. (C) Western blot for PDI or Erp5 expression in platelets at day 4 of control MO or 2 different MO. (D) Knockdown of PDI by a translational start site AUG-MO or 5' UTR-MO showed defects in thrombus formation after laser-injury compared with control MO in WT C57Bl/6J mice. Infusion of recombinant WT PDI and R120D mutant restored thrombus formation in the WT mice. Platelets, red; fibrin, green. Kinetic tracings of the median platelet (E) and fibrin (G) fluorescence (RFU) over time (sec).

Figure 5. Augmentation of thrombus formation in mice lacking GPx3. (A) Oxidative stress in $GPx3^{-/-}$ mice is prothrombotic after laser injury compared with WT mice. Platelets, red; fibrin, green. Kinetic tracings of the median (B) platelet and (C) fibrin fluorescence between WT and $GPx3^{-/-}$ mice. (D) Quantification of platelet fluorescence from panel B. (E) Quantification of fibrin fluorescence from panel C. Different colors represent injuries from different mice. (F) Kaplan-Meier analysis of the proportion of vessels unoccluded from panel G. (G) Intravital microscopy images of the $FeCl_3$ -induced carotid artery occlusion thrombosis model. Vessel autofluorescence and platelets are shown in green and red, respectively. N = 29 injuries from 3 different WT and N = 29 injuries from 3 different $GPx3^{-/-}$ mice in panels A-E. N = 5 different WT and $GPx3^{-/-}$ mice in panels F-G. * $P < .05$; **** $P < .0001$. AUC, area under the curve; RFU, relative fluorescence units.



platelet reactivity, and exacerbated vascular injury in disease models.^{25,26} When evaluated in the laser-induced vascular injury model of thrombus formation, median platelet accumulation was increased by 2.02-fold ($P = .038$) and median fibrin generation increased by 6.34-fold ($P < .001$) in the $GPx3^{-/-}$ mice compared with WT littermate controls (Figure 5A-E; supplemental Videos 12 and 13). Similarly, occlusion times following $FeCl_3$ exposure were accelerated in $GPx3^{-/-}$ mice compared with WT controls (Figure 5F-G). In contrast, bleeding times and prothrombin/activated partial thromboplastin times were unaffected (supplemental Figure 9A-D).

To assess the role of H_2O_2 -mediated sulfenylation of PDI in thrombus formation *in vivo*, we evaluated the ability of the sulfenylation-deficient variant, R120D PDI, to restore thrombus formation following PDI knockdown. The median platelet accumulation and fibrin generation in $GPx3^{-/-}$ mice

exposed to a PDI AUG-MO were decreased by 34% and 72%, respectively, compared with mice exposed to a control MO (Figure 6; supplemental Videos 14 and 15). As observed in WT mice, infusion of WT PDI reversed the defect in both platelet accumulation and fibrin formation (supplemental Video 16). Unlike native PDI and in WT mice, infusion of the R120D PDI mutant failed to reverse the defect in platelet accumulation caused by knockdown of PDI in $GPx3^{-/-}$ (Figure 6B; supplemental Video 17). Similarly, a PDI mutant lacking active site cysteines did not reverse the defect in platelet accumulation caused by knockdown of PDI (Figure 6B; supplemental Video 18). Evaluation of fibrin formation in $GPx3^{-/-}$ mice showed no significant differences in the effect of infusion of WT PDI, R120 mutant PDI, or catalytically dead PDI (Figure 6C). These results indicate that R120D is important in platelet accumulation in the presence, but not the absence (Figure 4), of oxidative stress.

Figure 4 (continued) (G) fluorescence showing rescue of thrombus formation by PDI knockdown with recombinant PDI infusion in WT C57Bl6J mice. (F) Quantification of platelet fluorescence from panel E. (H) Quantification of fibrin fluorescence from panel G. N = 30 injuries in control MO treatment; N = 31 injuries in PDI AUG-MO treatment; N = 30 injuries in PDI AUG-MO treatment with recombinant PDI infusion; N = 30 injuries in PDI AUG-MO treatment with PDI R120D infusion; N = 30 injuries in PDI 5' UTR MO treatment. In each cohort of injuries, the injuries were from 3 different mice. * $P < .05$; ** $P < .01$. Different colors represent injuries from different mice. AUC, area under the curve; Ctrl, control; ns, nonsignificant; RFU, relative fluorescence unit; Veh, vehicle.

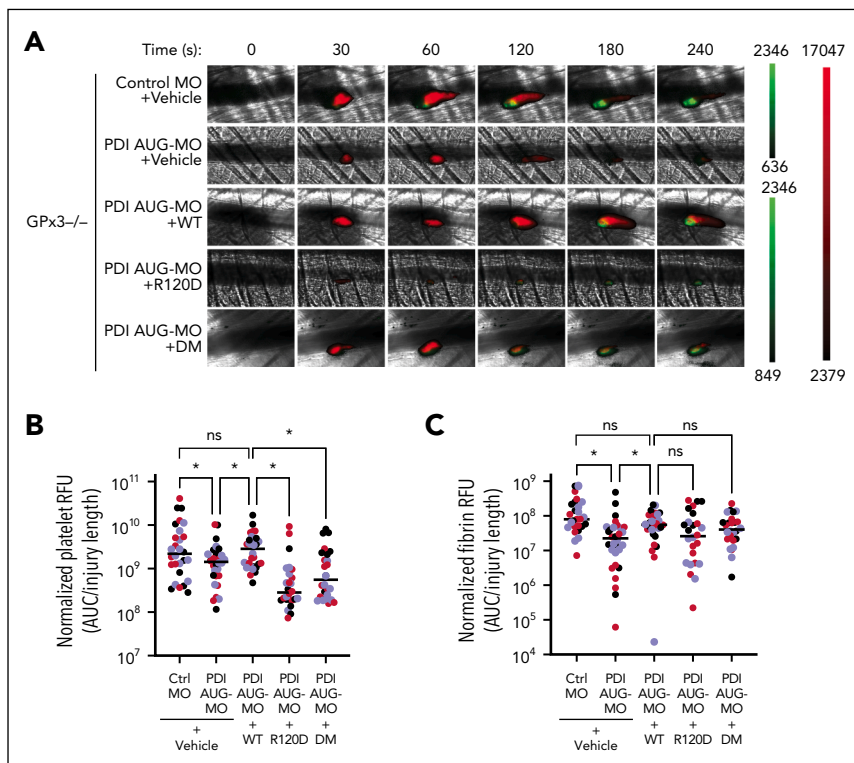


Figure 6. Sulfonylation-deficient R120D fails to restore augmented thrombosis in *GPx3*^{-/-} oxidative stress mice following MO-mediated PDI knockdown. (A) The prothrombotic phenotype of *GPx3*^{-/-} oxidative stress mice are prevented by knockdown of PDI (PDI AUG-MO) compared with control MO after laser injury to the cremaster arterioles. Sulfonylation-deficient R120D and the DM-PDI infusion are unable to rescue thrombus formation compared with WT PDI. Platelets, red; fibrin, green. (B) Quantification of platelet fluorescence from panel A. (C) Quantification of fibrin fluorescence from panel A. N = 30 injuries in PDI MO-treated *GPx3*^{-/-} mice with saline infusion; N = 28 injuries in control MO-treated *GPx3*^{-/-} mice with saline infusion; N = 30 injuries in PDI MO-treated *GPx3*^{-/-} mice with recombinant WT PDI infusion; N = 26 injuries in PDI MO-treated *GPx3*^{-/-} mice with recombinant PDI R120D infusion; N = 26 injuries in PDI MO-treated *GPx3*^{-/-} mice with recombinant DM PDI infusion. In each cohort of laser injuries, the injuries were from 3 different mice. Different colors represent injuries from different mice. **P* < .05. AUC, area under the curve; Ctrl, control; ns, nonsignificant; RFU, relative fluorescence units.

To further evaluate the role of PDI sulfonylation in platelet-mediated thrombosis, we tested the R120D PDI mutant in mice with a megakaryocyte/platelet-specific knockout of PDI (*PDI*^{fl/fl}:PF4 cre⁺). oxLDL infusion was used to promote thrombus formation by modeling oxidative lipid stress in dyslipidemia^{24,36,54-56} and we compared the ability of WT PDI, R120D PDI, and catalytically dead PDI to restore thrombus formation in mice lacking platelet PDI (Figure 7A). Occlusion times of carotid arteries in *PDI*^{fl/fl}:PF4 cre⁺ mice (ie, with platelet PDI) measured after induction of thrombus formation by FeCl₃ were significantly shortened from 11 ± 0.43 to 4.5 ± 0.56 minutes following infusion of oxLDL indicating enhanced thrombosis (Figure 7B, dashed black line; Figure 7C, open black circles). However, infusion of oxLDL in *PDI*^{fl/fl}:PF4 cre⁺ mice failed to reduce occlusion times (11 ± 1.2 minutes; Figure 7B, solid red line; Figure 7C, solid red circle). Infusion of WT PDI restored the effect of oxLDL on occlusion time in *PDI*^{fl/fl}:PF4 cre⁺ mice to 6.4 ± 0.69 minutes (Figure 7B, dashed red line; Figure 7C, open red circle). In contrast, infusion of R120D PDI failed to restore the decrease in occlusion time caused by oxLDL (11 ± 0.85 minutes; Figure 7B, dashed pink line; Figure 7C, open pink circle). Similarly, the catalytically dead mutant failed to restore oxLDL-mediated decrease in occlusion time (10 ± 1.2 minutes; Figure 7B, dashed gray line; Figure 7C, gray open circle). These results are consistent with the premise that sulfonylation of PDI is important for thrombus formation in oxidative stress.

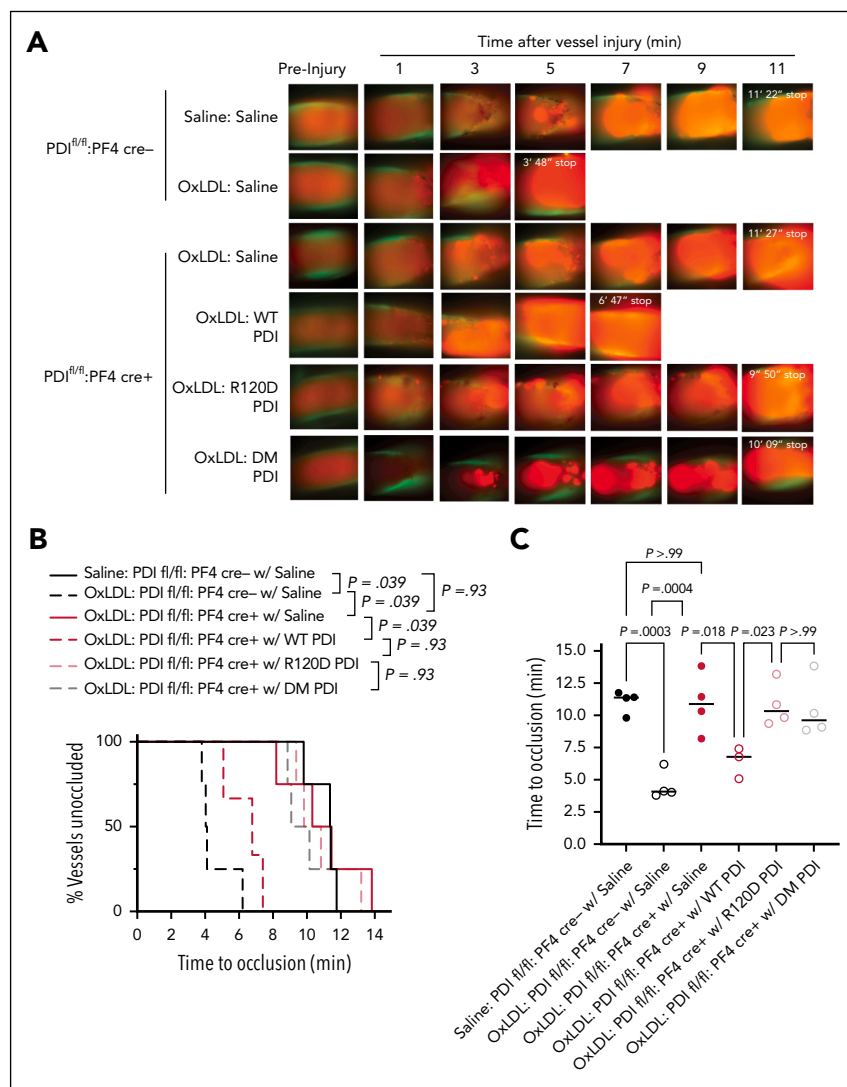
Discussion

Despite prior studies showing that oxidoreductase activity of PDI is essential for its role in thrombus formation in vivo,¹⁵ whether PDI acts primarily as an oxidase or a reductase

during thrombosis has previously been evaluated only indirectly. Trapping mutants of PDI have been used to identify PDI substrates in platelet-rich plasma, including vitronectin¹⁹ and histidine-rich glycoprotein, that form mixed disulfide bonds with PDI.¹⁸ PDI-mediated cleavage of these proteins enhances their activity. In addition, these proteins were shown to contribute to thrombus formation. Likewise, thrombospondin is a PDI substrate possessing a disulfide cleaved by PDI, exposing an Arg-Gly-Asp sequence.⁵⁷ Glycoprotein Ibα has also been shown to be cleaved by PDI, an action thought to promote thromboinflammation.⁵⁸ Challenges of these approaches include a bias for identification of PDI-mediated disulfide cleavage owing to detection methods and difficulty showing that a specific cleavage event is relevant during thrombus formation. In contrast, studies in platelets have shown that oxidized, but not reduced, PDI promotes platelet activation by oxidizing GSH to GSSG.⁵⁹ The possibility that PDI contributes to the oxidation of tissue factor has been proposed by several investigators.^{45,60-62} We have shown that oxidizing agents such as oxidized LDL particles can elicit the oxidation of PDI through a sulfonylated intermediate and that PDI contributes to thrombus formation in vivo under such oxidizing conditions.³⁶ All of these approaches, however, are indirect and leave the question whether PDI acts primarily as a reductase or an oxidase during thrombus formation, unanswered.

Our results now show that oxidation is a primary driver of PDI activity during thrombus formation in vivo. LOC14 was originally identified in a high throughput assay for its ability to inhibit PDI reductase activity.⁴³ We and others have identified small molecule inhibitors of PDI that block thrombus formation.^{11,44,63-66} What sets LOC14 apart is its ability to enhance oxidase and isomerase activities at the same time that it

Figure 7. Sulfenylation-deficient R120D fails to restore augmented thrombosis following oxLDL infusion in mice with platelet PDI deficiency. (A) oxLDL accelerated time to vessel occlusion after topical application of 10% FeCl₃ to the carotid arteries. Genetic deletion of PDI from platelets by the presence of platelet factor 4-cre prevented the accelerated time to vessel occlusion. The infusion of WT PDI, but not R120D or DM PDI, rescued the accelerated vessel occlusion time. Vessel autofluorescence, green; platelets, red. (B) Kaplan-Meier analysis of the proportion of vessels unoccluded from panel A. (C) Comparison of times to occlusion for individual mice. N = 4 mice in Saline: PDI^{fl/fl}:PF4 cre⁻ w/ Saline, OxLDL: PDI^{fl/fl}:PF4 cre⁻ w/ Saline, OxLDL: PDI^{fl/fl}:PF4 cre⁺ w/ Saline, OxLDL: PDI^{fl/fl}:PF4 cre⁺ w/ R120D PDI, and the OxLDL: PDI^{fl/fl}:PF4 cre⁺ w/ DM PDI cohort. N = 3 mice in the OxLDL: PDI^{fl/fl}:PF4 cre⁺ w/ WT PDI cohort.



inhibits reductase activity (Figure 1). It is therefore well suited for assessing whether oxidase or reductase activity of PDI is more important for in vivo thrombus formation. Evaluation in 2 independent models of thrombosis demonstrates that LOC14 augments thrombus formation (Figure 2), which we interpret as showing that PDI oxidase activity promotes thrombosis. Both substitution of the reactive sulfur with oxygen (as in ox-LOC14) and anti-PDI antibody reverse the augmenting activity of LOC14, thereby indicating that the ability of LOC14 to oxidize PDI is what renders it prothrombotic. We cannot rule out a role for PDI reductase activity in mechanisms of thrombosis that are not captured in our assays. Our in vivo observations, however, are consistent with in vitro studies which demonstrate that oxidized PDI promotes platelet aggregation.⁵⁹

Evaluation of the role of sulfenylation in thrombus formation demonstrated a previously unrecognized regulatory feature in PDI oxidation pathways in vivo. Following knockdown of endogenous PDI, a mutant PDI, R120D, that is incapable of sulfenylation was still able to coordinate thrombus formation in the absence of oxidative stress (Figure 4). This observation implies that sulfenylation is not required for thrombus

formation in the absence of oxidative stress. However, in GPx3^{-/-} mice, which have elevated baseline H₂O₂ levels owing to deficiency of a critical peroxidase,^{26,67} the R120D PDI mutant failed to restore platelet accumulation during thrombus formation (Figure 6). In mice lacking platelet PDI, WT PDI restored the reduction in occlusion time observed with infusion of oxLDL. Failure of the sulfenylation defective R120D PDI mutant to restore the decrease in occlusion time under the same conditions confirms a role for PDI sulfenylation in platelet accumulation during thrombus formation (Figure 7). The fact that the R120D PDI mutant can be oxidized by GSSG, but not sulfenylated via H₂O₂ (Figure 3), indicates distinct pathways for oxidation of PDI. These distinct PDI oxidation pathways may be active under different redox conditions, with PDI sulfenylation being relevant only in the setting of oxidative stress. The high prevalence of thrombosis in the setting of oxidative stress underscores the importance of this pathway of PDI oxidation.

Our structure-function studies of PDI sulfenylation show that sulfenylation of PDI by peroxides has mechanistic features resembling those involved in the reduction of H₂O₂ by peroxiredoxins. Unlike GPx family antioxidant enzymes that use a selenium active site and catalase that uses heme in the active

site, peroxiredoxins use a catalytic cysteine (the peroxidatic cysteine, C_p-SH). The thiolate of the peroxidatic cysteine (C_p-S-) reacts via nucleophilic attack with H₂O₂, which is held in place via hydrogen bonds with adjacent Arg and Thr backbones (supplemental Figure 4).^{47,48,68} This reaction results in H₂O and a cysteine sulfenic acid (C_p-SOH) intermediate that resolves with a second Cys (supplemental Figure 4).⁶⁹ The reaction is similarly facilitated by residues C56, R120, C53, and T101 in PDI, which are highly conserved among thiol isomerases (supplemental Figure 5). PDI sulfenylation by peroxides occurs at the “a” domain, and not the “a'” domain.³⁶ Within the catalytic CGHC motif of the “a” domain of PDI, the C53 thiolate typically has a lower acid dissociation constant (4.8-6.7) and initiates nucleophilic attack of substrates.²² The finding that C56, and not C53 (Figure 3), is sulfenylated was therefore unexpected. Previous studies evaluating the catalytic cycle of PDI have suggested that the acid dissociation constant of C56 can vary (8.6-10.5) depending on its proximity to R120.⁷⁰ We now show that R120 is also essential for peroxide-mediated sulfenylation. By analogy with the peroxiredoxins reaction center, we find that T101 also functions in this capacity. These side chains could stabilize the thiolate form of C56 and may help coordinate the peroxide with the catalytic motif. Further studies will be required to achieve an atomistic understanding of the reaction mechanism leading to PDI sulfenylation and its similarity to peroxiredoxins.

Limitations of this study include the fact that we have only evaluated arteriolar and arterial thrombosis. The issue of whether or not PDI functions in venous thrombus has not clearly been resolved. Similarly, the role of PDI in hemostasis remains somewhat controversial. Whether promotion of PDI activity may affect hemostasis in the setting of bleeding diatheses, however, remains largely unexplored.

Nonetheless, these findings not only clarify the distinct contributions of PDI redox states to thrombus formation *in vivo* but also underscore the broader significance of oxidative protein modifications in vascular biology. The discovery that the oxidative and/or isomerase activities of PDI drive its function during thrombus formation will affect the approaches used to identify PDI substrates and determine its prothrombotic functions. By elucidating a peroxide-driven sulfenylation mechanism involving conserved residues including a peroxidatic cysteine, we reveal a previously unappreciated mechanistic similarity between peroxiredoxins and thiol isomerases, demonstrating that PDI has peroxireductase activity. The fact that the triad (cysteine, arginine, and threonine) is conserved among thiol isomerases invoked in thrombus formation raises the possibility that other thiol isomerases may undergo sulfenylation under oxidative stress. Small molecules capable of inducing conformational changes that disrupt this mechanism of sulfenylation may have antithrombotic activity even in the presence of oxidative stress. In addition, carbon nucleophiles that covalently tag sulfenic acid to prevent further sulfur oxoform formation or allosterically disrupt protein-protein interactions may constitute effective antithrombotic approaches.⁷¹ Taken together,

these results underscore the importance of oxidized PDI in thrombosis and provide a mechanistic basis for how oxidative stress can modulate thrombus formation through specific posttranslational modifications.

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Authorship

Contribution: M.Y., N. Pozzi, and R.F. conceptualized the study and curated data; M.Y., O.C.O., M.C., N. Ponzar, and M.C.R. developed the methodology; M.Y., O.C.O., A.P., M.C., and N. Ponzar performed formal analysis; M.Y., O.C.O., A.P., J.T.F., Q.P.K., M.C., N. Ponzar, E.M.F., N. Pozzi, and R.F. performed investigation; M.C.R., A.B., K.S.C., D.E.H., J.L., and D.W.E. provided resources; M.Y. and R.F. wrote the original draft, reviewed and edited the manuscript, and contributed to supervision and project administration; M.Y. contributed to visualization; M.Y., A.P., D.W.E., N. Pozzi, and R.F. acquired funding; and all the authors read and agreed to the published version of the manuscript.

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Footnotes

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